

Topic Modeling McDonald's Reviews Case Study

DS 4002 – Spring 2026 – Siwen Liao

Due: see Canvas

Submission format: GitHub repository (submitted by link to Canvas)

Individual Assignment

General Description: Submit a link to your GitHub repository to Canvas.

Preparatory Assignments – None.

Why am I doing this? This case study provides you the opportunity to apply a popular topic modeling technique to a large real-world dataset, with over 33,000 reviews, that was scraped from Google. Restaurant reviews are common across various online platforms and understanding how to analyze them can result in a broader understanding of what consumers tend to focus on in reviews with respect to their sentiment. Ultimately, you will gain practical experience working with messy data, problem-solving skills to overcome conceptual and technical challenges, and experience with data science in business and human behavior applications.

What am I going to do? Your goal is to identify key themes in positive and negative reviews by applying the topic modeling approach LDA to uncover latent themes in a dataset of McDonald's reviews. You will use the resulting themes with the customer ratings to determine main themes based on sentiment. You will use contents in the provided GitHub repository to complete the case study, including the original dataset (not cleaned or preprocessed) of McDonald's reviews and a data appendix to reference if needed. You will go through the pipeline of writing code to clean and preprocess the dataset, produce a document term matrix, select the k (parameter) value for the model, fit the LDA model, export a dataset with the a LDA latent topic assigned to each review, and performing an analysis to find main themes based on sentiment. The goal by the end of this case study is to gain a thorough understanding of LDA as a topic modeling approach and apply your technical skills to a real-world business application.

GitHub repository link: https://github.com/siwenliao/DS4002_CS3/tree/main

Deliverables include:

- Cleaned and preprocessed dataset with a new column featuring LDA-produced topics
- Code that fits an LDA model with a selected k value and evaluation metrics
- Written document containing an analysis, interpretations, and discussions of your results
- GitHub repository containing data, code, and any additional materials relevant to the case study

Tips for success:

- Do not rush cleaning the data. The cleaning process is tedious and includes many necessary steps, such as removing punctuation, digits, and stop words, and performing lemmatization. However, this is arguably the most important step, and the better the data is cleaned, the higher the quality of the results will be.
- Use your own interpretation. Do not just simply trust the LDA model; make sure the themes make sense. If the themes are not humanly interpretable, consider adjusting the parameter (k) of the LDA model and prioritizing different evaluation metrics.
- Refer to the supplemental materials to help with your understanding.

How will I know I have succeeded? You will meet expectations on Topic Modeling McDonald's Reviews Case Study when you have followed the criteria in the rubric below.

Formatting	- One GitHub Repository containing all materials (submitted via link on Canvas) - The top-level page of the repository should contain: - A README.md file - A LICENSE.md (use MIT as default) - A SCRIPTS folder - A DATA folder - Written Analysis - References
README.md	<u>Goal</u>: This file serves as a high-level overview of your procedure. - Summarize the contents in your GitHub repository and your procedure for the case study, including any software and platform (e.g., Windows, Mac, Linux, etc.) used, to produce the resulting deliverables.
LICENSE.md	<u>Goal</u>: This file explains to a viewer the terms for when and how they may use your repository, including proper citation. - Select the MIT license when creating your GitHub repository.
SCRIPTS folder	<u>Goal</u>: This folder contains all the source code for the case study. - Include all scripts used. Name each script according to your procedure. - Include header comments at the beginning of a script to provide information of the contents of the script and what it achieves. - Include comprehensive and descriptive comments throughout your code so you can refer to it, and a viewer can follow your procedure easily. - You should have scripts that - Clean the data and perform preprocessing - Create a document term matrix - Choose an optimal k value for the LDA model - Assign topics and compare between different sentiments - Produce perplexity and coherence scores that evaluate the model
DATA Folder	<u>Goal</u>: This file should include the datasets used during the case study. - Include the final dataset with an additional column of a latent topic from the LDA model assigned to each review. - Include any intermittent datasets, if any, that you worked with.
Written Analysis PDF Document	<u>Goal</u>: This file should explain to a viewer your findings from the case study, including your interpretation and any applications of the results. - Have fun with this, as this is meant to be open-ended and for you to demonstrate what you have learned, how you and others can use your results (in daily life, as a business strategy, etc.), and demonstrate your work clearly to others (an important skill). The goal is about 1-3 pages.
References	- All references should be listed at the end of the document. - Use IEEE Documentation style (link).

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